AegisZero Web Application: Comprehensive UI & System Guide

Complete Screen-by-Screen, Button-by-Button & Component Walkthrough

Purpose of this Guide: This document provides an exhaustive, line-by-line explanation of every visual element, control button, KPI metric card, table entry, and multi-agent war-room tab featured in the AegisZero web application (`identity_app.py`). Use this as your master reference manual when explaining or demonstrating the live website to judges.

1. The Left Sidebar Navigation & Control Panel

The sidebar sits on the left of the screen and acts as the interactive mission control for the user or judge:

Sidebar Element	Function & Underlying Behavior	Visual Feedback on Screen
App Branding & Icon	Displays the shield insignia, app title ('AegisZero'), subtitle, and Track 1 metadata badge.	Establishes enterprise identity governance theme immediately.
■ Reset Directory	Resets all 5 employees back to their original unanalyzed state, wipes the swarm cache, and resets optimization state to False.	Restores all metrics to the un-optimized baseline (18% SOC2, 4 toxic keys, $55,500 waste).
■ Run 7-Agent Swarm (Primary Action)	The primary trigger. Ingests all employee profiles and passes them to the Master Orchestrator, which dispatches the 7 specialized agents in parallel and sequential order.	Shows an animated loading spinner ('7-Agent Swarm analyzing behavioral telemetry...'), followed by a green success banner and unlocks the War-Room.
■ 1-Click Autonomous Execution	Only appears *after* the Swarm has completed its analysis. Simulates zero-touch execution across cloud IAM and SCIM APIs with zero human tickets.	Fires celebratory digital balloons, turns toxic keys to 0, jumps SOC2 to 98%, and updates table status to '■ OPTIMIZED'.

2. The Top Executive KPI Metric Bar (5 Real-Time Cards)

Positioned at the top of the main dashboard, these 5 KPI metric cards give enterprise executives and judges an instantaneous summary of the health, financial waste, and security posture of the company:

Metric Card	Pre-Optimization State	Post-Optimization State	What It Proves to the Judges
1. Monitored Users	**5 Staff** (Active Directory)	**5 Staff** (Active Directory)	Confirms the sample directory size monitored across all departments.
2. Total SaaS Spend	**$7,186 / month** (Baseline spend)	**$7,186 / month** (Baseline spend)	Shows the current gross software subscription run-rate of the enterprise.
3. Annual Waste	**$55,500.00** -$4,625/mo bleed	**$55,500.00** Reclaimed! (100% ROI)	Direct monetary proof of FinOps cash recovery. Turns bleeding loss into reclaimed profit.
4. Toxic Admin Keys	**4 Keys** Critical Risk	**0 Keys** All Neutralized	Proves permanent high-privilege keys (AWS Root, Snowflake Admin) were revoked without incident.
5. SOC2 Compliance	**18% (Critical Fail)** Violations Active	**98% (Audit Ready)** +80% jump	Demonstrates immediate corporate audit readiness for SOC2 CC6.1 & ISO27001.

3. Enterprise Identity & Permission Risk Registry Table

This central table renders the full active directory. It shows how different departments and employee roles harbor hidden risks and waste. Here is the breakdown of every column and sample employee:

Table Columns Explained:

- **ID:** Unique employee enterprise badge identifier (e.g. EMP-101).
- **Name:** Employee full name.
- **Department:** Current organizational business unit.
- **Current Role:** Current active job title.
- **Role History:** Previous team assignment and the exact elapsed days since transition.
- **Active Apps:** Number of enterprise SaaS and cloud platforms currently provisioned to this user.
- **Monthly Spend:** Dollar sum of all active license subscriptions consumed by this user per month.
- **Zombie Apps:** Number of apps where user had 0 productive actions (commits/edits) for >30 days.
- **Admin Keys:** Count of permanent superuser/administrator credentials held by this employee.
- **Status Badge:** Dynamic status indicating compliance: ■ **CRITICAL ZOMBIE**, ■ **ZOMBIE DETECTED**, or ■ **OPTIMIZED**.

The 5 Pre-Configured Enterprise Personas in the Website:

Employee Persona	Department & Role Drift History	Licenses Held & Zombie Risk	Identified Annual Waste
1. David Chen (EMP-101)	Internal Tooling. *Former Payment Lead (left 75 days ago)*	AWS Root ($1200), Figma Pro ($75), GitHub Admin ($21), Datadog Admin ($180). **3 Zombie Apps + 3 Admin Keys**	**$17,700 / year** (Highest Risk Persona)
2. Sarah Jenkins (EMP-102)	Marketing & Growth. *Former Designer (switched 120 days ago)*	Figma Pro ($75), Salesforce Admin ($300), Miro Creator ($20). **2 Zombie Apps + 1 Admin Key**	**$2,940 / year**
3. Marcus Vance (EMP-103)	Customer Success. *Former Infra On-Call (switched 60 days ago)*	AWS Database Admin ($1200), PagerDuty Responder ($45), Zendesk Agent ($99). **2 Zombie Apps + 1 Admin Key**	**$14,940 / year**
4. Elena Rostova (EMP-104)	Mobile Engineering. *Senior iOS Dev (Stable, 0 days drift)*	GitHub Dev ($21), Figma Viewer ($0), Apple Org ($15). **0 Zombie Apps + 0 Admin Keys**	**$0.00 / year** (Compliant Control Baseline)
5. Tariq Al-Mansoor (EMP-105)	Data Science. *Former Cloud Contractor (transitioned 90d ago)*	Snowflake Accountadmin ($850), AWS Admin ($1200), GitHub Owner ($21). **3 Zombie Apps + 3 Admin Keys**	**$19,920 / year**

4. The Multi-Agent War-Room: Exhaustive Tab Walkthrough

When a user runs the swarm, the lower half of the website opens into the **Multi-Agent Autonomous War-Room**. A dropdown lets you select any employee (e.g., David Chen) to inspect their deliberations across 8 specialized tab views:

War-Room Tab	Visual Elements & Data Rendered	Technical Significance for Judges
Tab 1: ■ Master Orchestrator	• Terminal event bus stream with chronological timestamps. • Shows continuous ingestion, parallel dispatch to Agents 1-7, and consensus scoring.	Proves the central controller is not a hardcoded script, but a stateful DAG orchestrator.
Tab 2: ■ Agent 1: HR Context	• Assessment banner: *'Role shifted 75 days ago from Payment Gateway Lead to Internal Tooling'*. • Raw JSON telemetry displaying tenure metrics.	Proves the system cross-references HR organizational hierarchy with digital permissions.
Tab 3: ■ Agent 2: Behavioral Audit	• Success banner: *'Exposed 3 applications with active SSO logins but 0 productive contributions'*. • Table showing days since last login vs. days since last actual PR/Figma edit.	Demonstrates the first major breakthrough: exposing 'Passive Lurkers' whom Okta/Zylo miss.
Tab 4: ■■ Agent 3: Blast Radius	• Safety card: *'Scanned 42 CI/CD pipelines and cron jobs to ensure zero production crashes'*. • Score: **0.0% Blast Radius (VERIFIED SAFE TO REVOKE)**.	Demonstrates the second major breakthrough: eliminating IT fear of breaking production.
Tab 5: ■ Agent 4: FinOps Arbitrage	• Annual savings metric: **$17,700.00 / yr**. • Graceful Micro-Tiering plan table: from Tier & Cost to Target Downgrade Tier & Cost (e.g. AWS Root to JIT STS, Figma Pro to Free Viewer).	Proves concrete, auditable financial ROI that CFOs and IT directors care about.
Tab 6: ■ Agent 5: Security & SOC2	• Warning badge: **HIGH_AUDIT_FAIL**. • Exact breach citations: **SOC2 CC6.1 Principle of Least Privilege Violation** on AWS IAM.	Proves immediate compliance value by linking permissions to real cybersecurity standards.
Tab 7: ■ Agent 6: API Synthesizer	• Synthesized AWS CLI commands: `aws iam detach-user-policy ... && aws iam create-role --role-name JIT-STS-1hr`. • SCIM PATCH API curl payloads for Figma and Datadog.	Proves the system writes real, executable cloud configurations rather than just generating text.
Tab 8: ■ Agent 7: Slack Bot Comms	• Beautifully styled Slack message card addressed to David Chen. • Friendly explanation of graceful downgrade. • **1-Click Self-Healing Restore Link:** `https://aegiszero.internal/restore?user=EMP-101...`	Demonstrates the third major breakthrough: zero employee friction and zero IT helpdesk tickets.

5. Step-by-Step Live Demonstration Guide (For Presentation)

Follow this exact sequence of clicks during your live demo to deliver a seamless, high-impact presentation:

Demo Step	Action to Perform on Screen	What to Say to the Judges
Phase 1: Baseline View	Open the web app at `http://localhost:8501`. Point to the top KPI cards and the employee table.	*'Judges, notice our baseline: 5 employees, $7,186/mo spend, but $55,500/year bleeding in waste with an 18% failing SOC2 compliance score.'*
Phase 2: Trigger Swarm	Click the blue **'■ Run 7-Agent Swarm'** button in the left sidebar.	*'With one click, our Master Orchestrator dispatches our 7 specialized agents. Within 3 seconds, all 5 employees are audited across HR, Git, CI/CD, and CloudTrail.'*
Phase 3: Inspect Persona	Select **'David Chen'** in the dropdown. Click through Tabs 2, 3, 4, 6, and 8.	*'Look at David: HR confirms he left Payments 75 days ago; Behavioral audit proves 0 edits; Blast radius confirms 0% outage risk; and here is the synthesized AWS CLI command and friendly Slack card.'*
Phase 4: Autonomous Execution	Click the green **'■ 1-Click Autonomous Execution'** button in the sidebar.	*'Now watch zero-touch remediation: digital balloons fire, toxic admin keys drop to 0, SOC2 jumps from 18% to 98%, and we recover $55,500/year with zero human tickets!'*

6. Launch Command & Codebase Summary

The complete web application is built in Python using Streamlit, supported by modular simulation and multi-agent backend engines:

TERMINAL EXECUTION COMMAND:

```
cd C:\Users\DELL\.gemini\antigravity\scratch\aegismesh
streamlit run identity_app.py
```

CODE ARCHITECTURE FILE MAP:

- `identity_app.py`: Frontend web dashboard, KPI cards, table visualizer, tabbed war-room.
- `identity_engine.py`: Enterprise data models, employee personas, SaaS license metrics, simulation state.
- `identity_agents.py`: Master Zero-Trust Orchestrator + 7 specialized autonomous worker agents.